



UNIVERSITAS
GADJAH MADA

Marine Weather Decision System for Local Fishermen using C++

Wind Speed and Wave height Analysis for
Immediate Departure Decisions

*Merakyat, Mandiri, Berkelanjutan
Inclusive, Self-Reliant, Sustainable*





UNIVERSITAS
GADJAH MADA

GROUP 6 MEMBERS

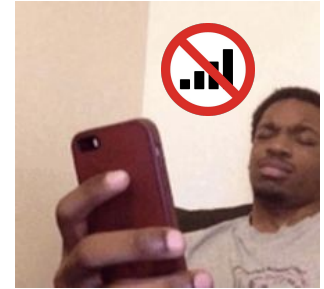
1. KEN ALFATHIR
2. ATHAYA DHIAN
3. ALBIZAR DIMAS

*Merakyat, Mandiri, Berkelanjutan
Inclusive, Self-Reliant, Sustainable*

ugm.ac.id

Problems

1. Local fishermen face sudden changes in marine weather.
2. Manual weather checking can lead to misjudgments and unsafe sea journeys.
3. Many coastal launch sites have zero cellular signal, rendering phone weather apps useless.



Solution



Provide an instant, automated safety recommendation whether to sail or not based on wind speed and wave height input.

Advantages Using C++

1. Designed to function in remote coastal zones with zero internet or cellular connectivity.
2. Lightweight and runs directly on low-power devices in the field.
3. Delivers instantaneous calculation without latency or heavy runtime dependencies.

Core logic & Algorithms for Sailing Warning

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      double wind_speed;
7      cout<< "Enter Wind_speed ";
8      cin>> wind_speed;
9
10     double wave_length;
11     cout<< "Enter Wave_length ";
12     cin>> wave_length;
13
14     if(wind_speed >= 34) {
15         cout << "STATUS: GALE WARNING! EXTREME HAZARD, DO NOT SAIL!";
16     }
17     else if (wind_speed >= 20 || wave_length >= 7.0) {
18         cout << "STATUS: SMALL CRAFT ADVISORY! HAZARDOUS FOR SMALL BOATS";
19     }
20     else {
21         cout << "STATUS: SAFE. condition are favorable for sailing";
22     }
23
24     return 0;
25 }
```

input

```
Enter Wind_speed 37
Enter Wave_length 7
STATUS: GALE WARNING! EXTREME HAZARD, DO NOT SAIL!

...Program finished with exit code 0
Press ENTER to exit console.
```

* Weather Warning criteria (e.g., Small Craft Advisories) adapted from National Weather Service Marine Definitions.

C++ Structures

- Input Parsing: Receives *Wind_speed* (in knots) and *Wave_height* (in feet) as numerical float variables.
- Conditional Branching (*if - else - if - else* structure):
 - ◆ Critical (Gale Warning) : Triggered if *Wind_speed* is more than 34 knots.
 - ◆ Warning (small craft advisory): Triggered if *Wind_speed* is more than 20 knots or *Wave_height* is more than 7 feet.
 - ◆ And safe if all the condition are below all hazard thresholds
- Output: Resulting the status code and actionable safety recommendation

Pseudocode

Program Weather Warning

```
BEGIN
    INPUT Wind_speed
    INPUT Wave_height

    IF Wind_speed >= 34
        OUTPUT "STATUS: GALE WARNING! Extreme hazard. Do not set sail."

    ELSE IF Wind_speed >= 20 and Wave_height >= 7
        OUTPUT "STATUS: SMALL CRAFT ADVISORY! HAZARDOUS FOR SMALL BOAT"

    ELSE
        OUTPUT "STATUS: SAFE. Conditions are favorable for sailing."
END
```

* Weather Warning criteria (e.g., Small Craft Advisories) adapted from National Weather Service Marine Definitions.

Thank You!

Project: Marine Weather Decision System for Local Fishermen (C++)

Presented by : Kelompok 6

References

National Weather Service. (n.d.). *Marine Definitions*. National Oceanic and Atmospheric Administration (NOAA).

https://www.weather.gov/key/marine_definitions

We would love to hear your feedback! Please feel free to ask any questions about our C++ logic, system thresholds, or anything else.